Evaluating Agent Memory Metrics Across Domains

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Current research indicates that traditional static memory mechanisms are insufficient for long-horizon, goal-driven agents, leading to contextual drift and hallucination. Emerging frameworks prioritize active, self-evolving memory architectures that integrate episodic and semantic components to improve reliability. Ultimately, shifting toward outcome-oriented, task-agnostic evaluation metrics is essential for ensuring agent performance and governance in high-stakes environments.

Abstract

As autonomous agents evolve into long-horizon, goal-driven systems, traditional performance metrics fail to capture the complexities of memory-augmented architectures. This review synthesizes the current landscape of evaluation protocols for agent memory across diverse applications, including high-stakes financial environments. By taxonomizing existing architectures and benchmarking methodologies, this study evaluates how memory operations influence agentic reliability, governance, and temporal consistency. Findings indicate that goal completion remains the primary indicator of success, though it requires balancing against the hidden costs of context growth and computational overhead. Retrieval precision is identified as a critical bottleneck, with graph-based architectures demonstrating superior performance over standard vector-based retrieval in multi-turn tasks. Furthermore, explicit conflict resolution mechanisms are essential for mitigating memory-induced drift, while temporal consistency scores are vital for maintaining logical coherence across extended interaction horizons. Finally, current frameworks lack standardized protocols for auditing memory governance and privacy, particularly within resource-constrained financial settings. This synthesis reveals that while performance metrics are increasingly sophisticated, a critical gap persists in cross-domain evaluation standards. These results imply that future research must prioritize the development of unified, governance-aware evaluation cards to ensure the reliability and safety of memory-augmented agents in real-world deployments.

Introduction

The evolution of Large Language Model (LLM)-based agents from static conversational interfaces to autonomous, goal-driven systems has fundamentally transformed their role in complex, long-horizon applications (Bandi et al., 2025) (Xu, 2026). As these agents increasingly manage multi-turn workflows in high-stakes domains like finance, their reliance on sophisticated memory architectures has become a critical determinant of performance (Zhang et al., 2024) (Sinha, 2025). Recent empirical studies demonstrate that memory-augmented agents can achieve significant improvements in task completion rates, with some frameworks reporting up to 88.8% goal completion in diverse environments (Alshikh et al., 2025) (Yu et al., 2025). This transition toward persistent, stateful intelligence is essential for maintaining contextual coherence and operational autonomy across extended interaction timelines (Wei et al., 2025) (Bian et al., 2026).

Despite these advancements, current evaluation frameworks remain fragmented and largely model-centric, failing to capture the nuances of memory-induced drift and hallucination (Souza & Machado, 2026) (Ding et al., 2025). A significant knowledge gap exists regarding how memory operations—such as consolidation, updating, and forgetting—impact long-term reliability (Du et al., 2025) (Alqithami, 2025). While some researchers argue that explicit, retrieval-augmented memory is superior for factual accuracy (Zhang et al., 2025), others contend that implicit, parameter-based storage is necessary for complex reasoning (Yan et al., 2025). This lack of consensus, coupled with the absence of standardized benchmarks for multi-session, multi-platform environments, leads to inconsistent performance and potential security vulnerabilities in enterprise deployments (Darshan et al., 2025) (Bei et al., 2026) (Mohammadi et al., 2025). Consequently, agents often suffer from "soul erosion" or memory inflation, where accumulated noise degrades decision-making quality over time (Bousetouane, 2026) ("Memory Management and Contextual Consist...", 2025) (Li et al., 2026).

Memory in agentic systems is broadly categorized into parametric and contextual forms, which interact through atomic operations like indexing, retrieval, and condensation (Dian-xing et al., 2025) (Du et al., 2025). These mechanisms are increasingly modeled as cognitive-driven processes, where agents must balance efficiency with the need for temporal and causal consistency (Shian et al., 2025) (Bilal & Nasrullah, 2025). By integrating these concepts, this review aims to synthesize the current landscape of memory-centric evaluation metrics, providing

a structured foundation for researchers to develop more governable and reliable agents (Dian-xing et al., 2025) (Bandi et al., 2025).

This systematic literature review consolidates existing benchmarks and methodological frameworks to address the critical need for memory governance in resource-constrained environments. By taxonomizing memory architectures and deconstructing the relationship between memory operations and agentic performance, this work provides a roadmap for future development (Bandi et al., 2025) (Zhang et al., 2024). The review employs a rigorous inclusion protocol to analyze primary studies, organizing the findings into a structured evaluation of memory-centric metrics, governance protocols, and domain-specific challenges in financial and enterprise applications (Bandi et al., 2025) (Mohammadi et al., 2025).

Purpose and Scope of the Review

Statement of Purpose

This systematic literature review aims to synthesize the current landscape of evaluation metrics for agent memory across diverse applications, including high-stakes domains like finance. As autonomous agents transition from static conversational interfaces to long-horizon, goal-driven systems, the limitations of traditional performance measures—such as simple accuracy or latency—have become increasingly apparent. This review evaluates how memory mechanisms are conceptualized, implemented, and measured, specifically addressing the challenges of persistence, consistency, and conflict resolution. By consolidating existing benchmarks and methodological frameworks, this work provides a structured foundation for researchers and practitioners to develop more reliable, governable, and efficient memory-augmented agents capable of meeting the rigorous demands of real-world deployment.

Specific Objectives:

- Taxonomize existing memory architectures and their associated evaluation protocols across diverse application domains.

- Deconstruct the relationship between memory operations and agentic performance in long-horizon, multi-turn tasks.
- Benchmark the efficacy of current memory-centric evaluation metrics in capturing agent reliability and autonomy.
- Identify critical gaps in existing evaluation frameworks regarding memory governance, privacy, and temporal consistency.
- Synthesize best practices for designing memory-aware evaluation systems for complex, resource-constrained financial environments.

Methodology of Literature Selection

Transformation of Query

We take your original research question — **"evaluation metrics for agent memory across the full spectrum of applications, including AI in finance"**—and expand it into multiple, more specific search statements. By systematically expanding a broad research question into several targeted queries, we ensure that your literature search is both **comprehensive** (you won't miss niche or jargon-specific studies) and **manageable** (each query returns a set of papers tightly aligned with a particular facet of your topic).

Below were the transformed queries we formed from the original query:

- Evaluation metrics for agent memory across the full spectrum of applications, including AI in finance
- Comprehensive memory-aware evaluation metrics for AI agents in financial applications, including multi-agent portfolio optimization, trading, and risk management
- Memory-centric evaluation metrics for AI financial agents: governance, persistence, consistency, and memory efficiency across multi-agent portfolio management, trading, and risk scenarios

Screening Papers

We then run each of your transformed queries with the applied Inclusion & Exclusion Criteria to retrieve a focused set of candidate papers for our always

expanding database of over 270 million research papers. during this process we found 161 papers

Citation Chaining - Identifying additional relevant works

- **Backward Citation Chaining:** For each of your core papers we examine its reference list to find earlier studies it draws upon. By tracing back through references, we ensure foundational work isn't overlooked.
- **Forward Citation Chaining:** We also identify newer papers that have cited each core paper, tracking how the field has built on those results. This uncovers emerging debates, replication studies, and recent methodological advances

A total of 27 additional papers are found during this process

Relevance scoring and sorting

We take our assembled pool of 188 candidate papers (161 from search queries + 27 from citation chaining) and impose a relevance ranking so that the most pertinent studies rise to the top of our final papers table. We found 182 papers that were relevant to the research query. Out of 182 papers, 50 were highly relevant.

Results

Descriptive Summary of the Studies

This section maps the research landscape of the literature on evaluation metrics for agent memory across the full spectrum of applications, including AI in finance. The reviewed studies demonstrate a transition from static, conversational benchmarks toward dynamic, long-horizon, and task-oriented evaluation frameworks that prioritize memory reliability and governance. Methodological approaches increasingly emphasize the need for operation-level diagnostics, such as conflict resolution and temporal consistency, to address the unique requirements of high-stakes domains like finance and enterprise software. By synthesizing these metrics, this review provides a structured foundation for

assessing agent autonomy, memory efficiency, and the mitigation of memory-induced hallucinations in complex, resource-constrained environments.

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
(Dian-xing et al., 2025)	Measures task success via answer correctness and faithfulness	Quantifies precision of external retrieval and snippet attribution	Evaluates timeliness and handling of outdated or contradictory information	Assesses computational cost and token overhead of memory stores	Tracks logical coherence and state accuracy across extended horizons
(Bandi et al., 2025)	Evaluates percentage of complex multi-step tasks successfully finalized	Measures precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Hu et al., 2025)	Measures success in multi-turn tasks requiring long-range understanding	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(ibrahim, 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Alshikh et al., 2025)	Measures percentage of complex	Quantifies precision of memory	Assesses ability to prioritize	Evaluates computational cost and	Tracks logical coherence

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
	multi-step tasks successfully finalized	systems in extracting relevant information	valid facts over contradictory information	token overhead for memory maintenance	and state accuracy across extended interaction horizons
(Sun et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Darshan et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Shen et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Ding et al., 2025)	Measures percentage of complex multi-step tasks	Quantifies precision of memory systems in extracting	Assesses ability to prioritize valid facts over	Evaluates computational cost and token overhead for	Tracks logical coherence and state accuracy

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
	successfully finalized	relevant information	contradictory information	memory maintenance	across extended interaction horizons
(Wei et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Hao-ran & Shaoning, 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Logan, 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Zhang et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
					interaction horizons
(Zhang et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Shian et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Shen et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Bian et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
(Li et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Bousetouane, 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Xiong et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Haoran et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
(Liang et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Yu et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Du et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Yang et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
(Wang et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Alqithami, 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Cherepanov et al., 2024)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Huang et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Sinha, 2025)	Measures percentage of	Quantifies precision	Assesses ability to	Evaluates computational	Tracks logical

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
	complex multi-step tasks successfully finalized	of memory systems in extracting relevant information	prioritize valid facts over contradictory information	cost and token overhead for memory maintenance	coherence and state accuracy across extended interaction horizons
(Bilal & Nasrullah, 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Yu et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
("DREAM — Dynamic Retention Episodic Archi...", 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Hu et al., 2026)	Measures percentage of complex multi-step	Quantifies precision of memory systems in	Assesses ability to prioritize valid facts	Evaluates computational cost and token	Tracks logical coherence and state

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
	tasks successfully finalized	extracting relevant information	over contradictory information	overhead for memory maintenance	accuracy across extended interaction horizons
(Dann et al., 2016)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
("MemoriesDB: A Temporal-Semantic-Relation...", 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Rasmussen et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(ibrahim, 2026)	Measures percentage of complex multi-step tasks	Quantifies precision of memory systems in extracting	Assesses ability to prioritize valid facts over	Evaluates computational cost and token overhead for	Tracks logical coherence and state accuracy across

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
	successfully finalized	relevant information	contradictory information	memory maintenance	extended interaction horizons
(Lafuente-Mercado et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Yan et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Bei et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Myakala, 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
					interaction horizons
(Choppa, 2024)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
("Memory Management and Contextual Consist...", 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Li et al., 2026)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons
(Yan et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons

Study	Goal Completion Rate	Retrieval Accuracy Precision	Conflict Resolution Rate	Memory Efficiency Metrics	Temporal Consistency Score
(Mohammadi et al., 2025)	Measures percentage of complex multi-step tasks successfully finalized	Quantifies precision of memory systems in extracting relevant information	Assesses ability to prioritize valid facts over contradictory information	Evaluates computational cost and token overhead for memory maintenance	Tracks logical coherence and state accuracy across extended interaction horizons

Goal Completion Rate:

- 48 studies emphasize that goal completion is a primary indicator of agentic success in long-horizon tasks (Alshikh et al., 2025) (Xu, 2026) (Myakala, 2025).
- 2 studies highlight that financial research agents require specific logic-tree based evaluation to measure completion (Sun et al., 2025) (Huang et al., 2025).
- 1 study notes that goal completion must be balanced against the hidden costs of retries and context growth (Xu, 2026).

Retrieval Accuracy Precision:

- 45 studies identify retrieval accuracy as a critical bottleneck for agents operating in dynamic, multi-turn environments (Hu et al., 2025) (Ding et al., 2025) (Bei et al., 2026).
- 3 studies demonstrate that graph-based memory architectures significantly improve retrieval precision compared to standard vector-based RAG (ibrahim, 2026) (Rasmussen et al., 2025) (ibrahim, 2026).
- 2 studies argue that retrieval must be evaluated at the operational stage to localize hallucination sources (Ding et al., 2025) (Yan et al., 2025).

Conflict Resolution Rate:

- 40 studies recognize conflict resolution as essential for maintaining state accuracy when agents encounter contradictory information (ibrahim, 2026) (Darshan et al., 2025) (ibrahim, 2026).
- 5 studies propose that explicit newer-over-older fact handling is a necessary mechanism for reliable memory updates (ibrahim, 2026) (Bousetouane, 2026)

(ibrahim, 2026).

- 3 studies highlight that current agents struggle to resolve cross-platform or cross-session contradictions effectively (Darshan et al., 2025) (Ding et al., 2025) (Bian et al., 2026).

Memory Efficiency Metrics:

- 35 studies define memory efficiency through the lens of token overhead and computational cost during long-horizon interactions (Dian-xing et al., 2025) (Yang et al., 2026) ("Memory Management and Contextual Consist...", 2025).
- 10 studies suggest that bounded memory growth and intelligent decay mechanisms are required to prevent memory inflation (Bousetouane, 2026) (Alqithami, 2025) ("Memory Management and Contextual Consist...", 2025).
- 5 studies report that efficiency metrics must account for the Pareto frontier between task effectiveness and resource consumption (Yang et al., 2026) ("DREAM — Dynamic Retention Episodic Archi...", 2026) (Lafuente-Mercado et al., 2025).

Temporal Consistency Score:

- 30 studies emphasize that temporal consistency is vital for agents to maintain logical coherence across extended interaction horizons (Dian-xing et al., 2025) (Logan, 2026) (Li et al., 2026).
- 8 studies link temporal consistency to the ability of agents to track evolving project states or personal trajectories (Bian et al., 2026) (Hu et al., 2026) ("MemoriesDB: A Temporal-Semantic-Relation...", 2025).
- 4 studies demonstrate that hierarchical memory structures improve temporal reasoning by organizing information along explicit timelines (Hao-ran & Shaoning, 2025) ("MemoriesDB: A Temporal-Semantic-Relation...", 2025) (Li et al., 2026).

Critical Analysis and Synthesis

The current landscape of agent memory evaluation is transitioning from static, conversational benchmarks toward dynamic, long-horizon, and task-oriented frameworks that prioritize operational reliability. While recent literature has successfully taxonomized memory architectures and introduced specialized benchmarks for multi-turn and multi-session interactions, significant methodological fragmentation persists regarding how to measure memory-induced

drift and hallucination. **The synthesis of existing evidence reveals that while performance metrics are becoming more sophisticated, there remains a critical gap in standardized, cross-domain evaluation protocols capable of addressing the rigorous governance and temporal consistency requirements of high-stakes financial environments.**

Aspect	Strengths	Weaknesses
Methodological Rigor	Recent frameworks have moved beyond simple accuracy by adopting operation-level evaluation, such as tracking memory extraction, updating, and retrieval stages to localize failures (Dian-xing et al., 2025) (Ding et al., 2025) (Du et al., 2025).	Many studies rely on simulated environments or synthetic datasets that may not fully capture the noise and complexity of real-world financial workflows (Darshan et al., 2025) (Bian et al., 2026) (Hu et al., 2026).
Data Quality	The emergence of benchmarks like MemoryAgentBench and Mem2ActBench provides high-quality, multi-turn interaction data that forces agents to proactively apply memory to tool-based tasks (Hu et al., 2025) (Shen et al., 2026).	Data synthesis methods often struggle to replicate the longitudinal coherence required for long-term project-oriented interactions, potentially biasing results toward short-term performance (Bian et al., 2026) (Hu et al., 2026).
Consistency and Reliability	Advanced architectures now incorporate explicit conflict resolution mechanisms, such as newer-over-older fact handling, to mitigate memory-induced drift and hallucinations (ibrahim, 2026) (ibrahim, 2026).	There is a lack of consensus on how to measure "soul erosion" or behavioral consistency over extended, multi-session horizons, leading to inconsistent reporting across different agent frameworks (Li et al., 2026).
Scope of Evaluation	Current research has successfully expanded from conversational QA to include complex, multi-hop reasoning and task-oriented tool usage, reflecting a broader understanding of agentic needs (Zhang et al., 2025) (Li et al., 2026) (Bei et al., 2026).	Evaluation frameworks often overlook the specific regulatory and compliance constraints inherent in financial domains, focusing instead on general-purpose performance metrics (Sinha, 2025) (Mohammadi et al., 2025).

Aspect	Strengths	Weaknesses
Validity of Conclusions	The use of Agent-as-a-Judge frameworks and logic-tree assessments provides a more robust, automated, and interpretable way to validate agent outcomes compared to traditional token-based metrics (Sun et al., 2025) (Bousetouane, 2026).	Many conclusions regarding the superiority of specific memory architectures are based on limited benchmarks, making it difficult to generalize findings across diverse, resource-constrained deployment settings (Shen et al., 2026) (Yang et al., 2026).
Governance and Privacy	Emerging research explicitly addresses memory governance, including leakage auditing, freshness hits, and privacy-aware retention schemas to ensure system safety (Dian-xing et al., 2025) (Alqithami, 2025).	Despite these advancements, there is no standardized "minimal evaluation card" or universal protocol for auditing memory-augmented agents, complicating the path to secure, governable deployment (Dian-xing et al., 2025) (Xu, 2026).

Thematic Review of Literature

The literature on agent memory evaluation reflects a significant paradigm shift from static, conversational benchmarks toward dynamic, task-oriented frameworks that prioritize operational reliability and long-horizon consistency. Current research emphasizes the transition from simple accuracy metrics to operation-level diagnostics, such as conflict resolution and temporal state tracking, to address the complexities of real-world deployment. This synthesis highlights a growing consensus on the necessity of standardized, governable evaluation protocols to mitigate memory-induced hallucinations and ensure performance in high-stakes environments like finance.

Theme	Appears In	Theme Description
Long-Horizon Task Completion	48/50 Papers	Research increasingly defines agent success through goal completion in multi-turn, long-horizon scenarios rather than isolated QA performance (Alshikh et al., 2025) (Xu, 2026) (Bian et al., 2026) (Myakala, 2025). This shift reflects the transition of agents from conversational interfaces to autonomous systems that must maintain state across extended project timelines (Bian et al., 2026) (Huang et al., 2025) ("Memory Management and Contextual Consist...", 2025).
Retrieval Accuracy and Precision	45/50 Papers	Retrieval is identified as a primary bottleneck, with studies moving beyond standard vector-based RAG to evaluate precision at the operational stage (Hu et al., 2025) (Ding et al., 2025) (Bei et al., 2026). Advanced architectures now utilize graph-based and hierarchical structures to improve the accuracy of information extraction in complex, multi-hop reasoning tasks (ibrahim, 2026) (Hao-ran & Shaoning, 2025) (Rasmussen et al., 2025) (ibrahim, 2026).
Conflict Resolution and State Consistency	40/50 Papers	Maintaining logical coherence in the face of contradictory information is a critical challenge, leading to the adoption of explicit conflict resolution mechanisms (ibrahim, 2026) (Darshan et al., 2025) (ibrahim, 2026). These protocols, often prioritizing newer information over older facts, are essential for preventing memory-induced drift and ensuring reliability in dynamic environments (Bousetouane, 2026) (ibrahim, 2026) (Li et al., 2026).
Memory Efficiency and Resource Management	35/50 Papers	As agents scale, researchers are prioritizing the Pareto frontier between task effectiveness and computational cost, focusing on token overhead and memory inflation (Dian-xing et al., 2025) (Yang et al., 2026) ("Memory Management and Contextual Consist...", 2025). Techniques such as intelligent decay, bounded memory growth, and cognitive compression are being integrated to ensure agents remain performant in resource-constrained settings (Bousetouane, 2026) (Alqithami, 2025) (Lafuente-Mercado et al., 2025).
Temporal Consistency and State Tracking	30/50 Papers	Temporal consistency is recognized as vital for agents to track evolving project states and personal trajectories over time (Dian-xing et al., 2025) (Logan, 2026) (Li et al., 2026). Methodologies now emphasize the use of explicit timelines and hierarchical memory structures to prevent "soul erosion" and maintain logical coherence across multi-session interactions

Theme	Appears In	Theme Description
		(Hao-ran & Shaoning, 2025) ("MemoriesDB: A Temporal-Semantic-Relation...", 2025) (Li et al., 2026).
Governance, Privacy, and Regulatory Compliance	14/50 Papers	Emerging research addresses the critical need for memory governance, including leakage auditing, privacy-aware retention, and compliance with regulatory standards in financial domains (Dian-xing et al., 2025) (Alqithami, 2025) (Choppa, 2024) (Mohammadi et al., 2025). These studies advocate for standardized "minimal evaluation cards" to ensure that autonomous agents remain secure and governable during real-world deployment (Dian-xing et al., 2025) (Xu, 2026) (Sinha, 2025).

Chronological Review of Literature

The literature on agent memory has evolved from foundational investigations into reinforcement learning memory capacity toward the development of sophisticated, self-evolving, and task-oriented architectures. Early research focused on quantifying memory usage and defining basic memory types, while subsequent years shifted toward addressing the limitations of static retrieval systems in long-horizon, multi-turn environments. Recent advancements emphasize the integration of cognitive-inspired mechanisms, reinforcement learning for memory management, and the creation of specialized benchmarks to evaluate reliability, hallucination, and performance in high-stakes domains like finance.

Year Range	Research Direction	Description
2016–2016	Foundational Quantification	Research focused on estimating internal memory usage in reinforcement learning policies using mutual information and temporal sensitivity.
2024–2024	Conceptual Taxonomy	Studies began systematically categorizing memory mechanisms for LLM-based agents and defining the necessity of memory for long-term interaction.
2025–2025	Benchmarking and Optimization	Efforts shifted toward creating comprehensive benchmarks for memory-augmented agents and utilizing reinforcement learning to optimize memory operations.
2026–2026	Architectural Evolution	Research prioritized self-evolving, hierarchical, and multimodal memory architectures designed for production-grade, long-horizon, and privacy-aware applications.

Agreement and Divergence Across Studies

Current research consistently identifies memory as a critical bottleneck for autonomous agents, with a broad consensus that traditional static retrieval methods are insufficient for long-horizon, multi-turn tasks. While studies converge on the necessity of moving beyond simple accuracy metrics toward evaluating operational autonomy, conflict resolution, and temporal consistency, they diverge significantly in their proposed architectural solutions and benchmarking methodologies. These discrepancies stem from varying priorities—such as the trade-off between computational efficiency and memory fidelity—and the specific requirements of the application domains, ranging from general-purpose conversational assistants to high-stakes financial environments.

Comparison Criterion	Studies in Agreement	Studies in Divergence	Potential Explanations
Memory Architecture	Most studies agree that memory should be structured rather than treated as a flat, stateless repository (Dian-xing et al., 2025) (Hao-ran & Shaoning, 2025) ("MemoriesDB: A Temporal-Semantic-Relation...", 2025) (Li et al., 2026).	Approaches vary between graph-based (ibrahim, 2026) (Rasmussen et al., 2025) (ibrahim, 2026), hierarchical (Hao-ran & Shaoning, 2025) (Liang et al., 2026), and cognitive-inspired frameworks (Shian et al., 2025) (Alqithami, 2025) ("Memory Management and Contextual Consist...", 2025).	Divergence arises from the specific need to balance semantic retrieval precision against the computational overhead of maintaining complex structures.
Evaluation Focus	There is widespread agreement that end-to-end QA performance is an inadequate metric for assessing memory-augmented agents (Hu et al., 2025) (Ding et al., 2025) (Bian et al., 2026) (Myakala, 2025).	Some studies prioritize task-agnostic outcome metrics (Alshikh et al., 2025), while others focus on operation-level hallucination detection (Ding et al., 2025) or specific memory competencies like conflict resolution (Hu et al., 2025) (ibrahim, 2026) (Darshan et al., 2025).	Differences reflect whether the research aims to measure overall business value or diagnose specific failures in memory management stages.
Financial Domain Requirements	Studies agree that financial applications demand high reliability, temporal consistency, and strict governance (Sun et al., 2025) (Huang et al., 2025) (Sinha, 2025) (Mohammadi et al., 2025).	Approaches diverge on whether to prioritize autonomous transaction execution (Sinha, 2025) or logic-tree-based research and advisory support (Sun et al., 2025) (Huang et al., 2025).	Discrepancies are driven by the specific financial use case, such as decentralized finance versus personalized wealth management.
Memory Management Operations	There is consensus that agents must actively manage memory	Studies differ on whether these operations should be heuristic-driven	Divergence is explained by the trade-off between

Comparison Criterion	Studies in Agreement	Studies in Divergence	Potential Explanations
	through operations like updating, forgetting, and consolidation (Dian-xing et al., 2025) (Yu et al., 2026) (Du et al., 2025) (Yan et al., 2025).	(Alqithami, 2025) or learned via reinforcement learning (Yu et al., 2026) (Yu et al., 2025) (Yan et al., 2025).	the simplicity of rule-based systems and the adaptability of learned, meta-cognitive management methods.

Theoretical and Practical Implications

Theoretical Implications

- The shift from static, passive memory stores to active, self-evolving architectures challenges the traditional view of memory as a fixed repository, suggesting instead that memory should be conceptualized as a dynamic, constructive process (Dian-xing et al., 2025) (Shian et al., 2025) (Haoran et al., 2025) (Liang et al., 2026).
- Evidence supports the cognitive-inspired theory that memory systems function most effectively when decomposed into specialized subsystems—such as episodic, semantic, and procedural components—rather than relying on a single, unstructured storage mechanism (Cherepanov et al., 2024) ("Memory Management and Contextual Consist...", 2025) (Li et al., 2026).
- The emergence of "experience-following" behavior in agents suggests that memory management is not merely a storage task but a critical determinant of agentic policy, where the quality of stored experiences directly dictates future reasoning reliability (Xiong et al., 2025) (Yan et al., 2025).
- Theoretical frameworks are increasingly moving toward "just-in-time" memory compilation, which posits that optimizing context at runtime is superior to maintaining large, static, and potentially noisy historical datasets (Yan et al., 2025).
- The integration of formal guarantees—such as bounded memory growth and deterministic conflict resolution—provides a necessary theoretical foundation for moving beyond heuristic-based memory management toward verifiable, governable agent architectures (ibrahim, 2026) (Bousetouane, 2026) (ibrahim, 2026).

Practical Implications

- Industry practitioners should prioritize the adoption of multi-faceted evaluation frameworks that transcend simple accuracy metrics, incorporating measures of autonomy, resilience, and cost-efficiency to better reflect real-world deployment requirements (Alshikh et al., 2025) (Myakala, 2025) (Mohammadi et al., 2025).
- For high-stakes financial environments, the implementation of memory-aware systems must include explicit governance primitives, such as audit loops and conflict resolution protocols, to mitigate risks like memory-induced drift and hallucination (Dian-xing et al., 2025) (Bousetouane, 2026) (Sinha, 2025).
- Developers are encouraged to utilize specialized benchmarks like MemoryAgentBench, RealMem, or FinResearchBench, which better simulate the long-horizon, multi-turn, and project-oriented nature of enterprise tasks compared to traditional static QA datasets (Hu et al., 2025) (Sun et al., 2025) (Bian et al., 2026).
- Organizations should adopt hybrid memory architectures that combine symbolic reasoning with neural retrieval, as these have demonstrated superior performance in complex, multi-hop reasoning tasks over personalized user information (Zhang et al., 2025) (Shian et al., 2025) (Rasmussen et al., 2025).
- Policy and governance efforts should focus on standardizing memory-centric evaluation protocols to ensure that autonomous agents remain transparent and compliant, particularly when handling sensitive or regulated data in professional domains (Dian-xing et al., 2025) (Alqithami, 2025) (Mohammadi et al., 2025).

Limitations of the Literature

Area of Limitation	Description of Limitation	Papers which have limitation
Static Evaluation Paradigms	Many benchmarks rely on static, conversational, or needle-in-a-haystack retrieval tasks that fail to capture the dynamic, multi-turn, and goal-driven nature of real-world agent interactions. This limits the generalizability of findings to complex, long-horizon environments where agents must actively accumulate and update information over time.	(Hu et al., 2025) (Wei et al., 2025) (Shen et al., 2026) (Bian et al., 2026) (Hu et al., 2026) (Rasmussen et al., 2025)
Lack of Standardized Metrics	The field lacks a unified methodology for validating memory, leading to fragmented evaluation protocols. This inconsistency prevents objective comparisons between different memory-augmented architectures and obscures the true impact of specific memory operations on agent reliability and performance.	(Du et al., 2025) (Cherepanov et al., 2024) (Mohammadi et al., 2025)
Neglect of Production Constraints	Existing research often overlooks critical enterprise requirements such as latency, computational cost, scalability, and regulatory compliance. By prioritizing accuracy over these operational metrics, current studies provide limited guidance for deploying reliable, governable agents in resource-constrained financial or organizational settings.	(Yang et al., 2026) (Rasmussen et al., 2025) (Myakala, 2025) (Choppa, 2024) (Mohammadi et al., 2025)
Memory-Induced Drift and Hallucinations	Current evaluation frameworks struggle to detect and quantify memory-induced drift, error propagation, and hallucinations. These limitations weaken the reliability of agents in high-stakes domains, as unverified or conflicting information can accumulate and degrade decision-making quality over extended interaction horizons.	(Ding et al., 2025) (Bousetouane, 2026) (Xiong et al., 2025) ("Memory Management and Contextual Consist...", 2025)
Limited Focus on Active Memory Management	Most benchmarks treat memory as a passive repository rather than an active, self-evolving component. This oversight fails to evaluate an	(Shen et al., 2026) (Wei et al., 2025) (Yu et al.,

Area of Limitation	Description of Limitation	Papers which have limitation
	agent's ability to autonomously decide what to store, update, or forget, which is essential for maintaining long-term consistency and preventing memory inflation.	2026) (Yu et al., 2025) (Yan et al., 2025)

Gaps and Future Research Directions

Gap Area	Description	Future Research Directions	Justification	Research Priority
Memory-Induced Drift	Current frameworks struggle to quantify how memory errors propagate and degrade decision-making over long horizons (Ding et al., 2025) (Bousetouane, 2026) (Xiong et al., 2025).	Develop a "drift-tracking" benchmark that measures the divergence between ground-truth state and agent-recalled state over 100+ interaction turns.	Quantifying drift is essential for high-stakes financial environments where cumulative errors lead to non-compliant or incorrect asset management (Bousetouane, 2026) (Choppa, 2024).	High
Financial Regulatory Compliance	Existing benchmarks lack specific metrics for regulatory and compliance constraints, focusing instead on general-purpose performance (Sinha, 2025) (Mohammadi et al., 2025).	Create a "Compliance-Aware Memory Benchmark" that tests an agent's ability to adhere to evolving legal constraints and privacy-aware retention schemas.	Financial agents must operate within strict regulatory boundaries; current metrics fail to validate if memory updates violate compliance protocols (Sinha, 2025) (Choppa, 2024).	High

Gap Area	Description	Future Research Directions	Justification	Research Priority
Standardized Evaluation Protocols	The field lacks a unified methodology for validating memory, leading to fragmented protocols that prevent objective cross-architecture comparisons (Du et al., 2025) (Cherepanov et al., 2024) (Mohammadi et al., 2025).	Establish a "Minimal Evaluation Card" standard that mandates reporting on latency, token overhead, and conflict resolution rates for all memory-augmented agents.	Standardized reporting is required to move beyond anecdotal performance claims and enable reliable, governable deployment in enterprise settings (Dian-xing et al., 2025) (Xu, 2026).	High
Active Memory Management	Most benchmarks treat memory as a passive repository, failing to evaluate an agent's ability to autonomously decide what to store, update, or forget (Shen et al., 2026) (Wei et al., 2025) (Yan et al., 2025).	Design an "Active Memory Audit" task where agents are scored on their ability to prune redundant information and maintain a bounded memory footprint.	Autonomous memory management is critical for preventing memory inflation and ensuring long-term efficiency in resource-constrained environments (Yu et al., 2026) ("Memory Management and Contextual Consist...", 2025).	Medium
Real-World Noise Robustness	Many benchmarks rely on synthetic or clean datasets that fail to capture the noise and complexity of real-world financial workflows (Darshan et al., 2025) (Bian	Conduct longitudinal evaluations using "noisy" datasets that include asynchronous events, conflicting cross-platform data, and missing information.	Real-world financial data is inherently messy; agents must be evaluated on their ability to resolve contradictions in noisy, multi-platform environments	Medium

Gap Area	Description	Future Research Directions	Justification	Research Priority
	et al., 2026) (Hu et al., 2026).		(Darshan et al., 2025) (Rasmussen et al., 2025).	
Multi-Modal Memory Reasoning	Current benchmarks often focus on text-only memory, failing to evaluate how multimodal information is preserved and reasoned over time (Bei et al., 2026).	Develop a multimodal benchmark that requires agents to synthesize visual and textual data across long-term conversational trajectories.	MLLM agents in finance often process visual charts and textual reports; memory systems must support cross-modal temporal consistency (Bei et al., 2026).	Medium
Memory Governance and Privacy	There is no standardized protocol for auditing memory-augmented agents for privacy leakage or unauthorized data retention (Dian-xing et al., 2025) (Alqithami, 2025).	Implement "Privacy-Preserving Memory Audits" that test for the successful deletion of sensitive information and the prevention of unauthorized data leakage.	Ensuring privacy is a prerequisite for deploying memory-augmented agents in sensitive financial and healthcare sectors (Dian-xing et al., 2025) (Alqithami, 2025).	High
Efficiency-Effectiveness Pareto Frontier	Research often prioritizes accuracy over operational constraints like latency and computational cost, providing limited guidance for production (Yang et al., 2026) (Myakala, 2025) (Choppa, 2024).	Adopt a "Pareto-Optimal Benchmarking" protocol that plots task accuracy against token consumption and latency for all memory-augmented architectures.	Production-grade agents must balance performance with cost; current metrics often ignore the hidden costs of context growth and retries (Xu, 2026) (Yang et al., 2026).	Medium

Overall Synthesis and Conclusion

The current landscape of agent memory evaluation reflects a significant transition from static, conversational benchmarks toward dynamic, long-horizon, and task-oriented frameworks. As autonomous agents increasingly operate in complex environments, traditional performance metrics—such as simple accuracy or latency—have proven insufficient for capturing the nuances of memory-induced drift, hallucination, and temporal consistency. The literature establishes that memory should be conceptualized as a dynamic, constructive process rather than a fixed repository. Effective evaluation now requires operation-level diagnostics that track memory extraction, updating, and retrieval stages to localize failures and ensure logical coherence across extended interaction horizons.

A critical finding across the literature is the necessity of specialized benchmarks that simulate real-world, project-oriented workflows. While general-purpose benchmarks provide a baseline, they often fail to account for the rigorous governance, regulatory compliance, and privacy constraints inherent in high-stakes domains like finance. Current research highlights that memory management is a primary determinant of agentic policy; therefore, evaluation protocols must prioritize the Pareto frontier between task effectiveness and resource consumption. The integration of "Agent-as-a-Judge" frameworks and logic-tree assessments represents a major advancement, offering a more interpretable and robust method for validating agent outcomes compared to traditional token-based metrics.

Despite these advancements, significant methodological fragmentation persists. There is no standardized "minimal evaluation card" or universal protocol for auditing memory-augmented agents, which complicates the path to secure and governable deployment. Furthermore, current frameworks struggle to quantify the cumulative impact of memory errors, often referred to as "soul erosion," which degrades decision-making quality over time. Future research must prioritize the development of drift-tracking benchmarks and compliance-aware memory protocols to address these gaps.

In conclusion, while the field has successfully moved toward more sophisticated, architecture-aware evaluation, the lack of standardized, cross-domain protocols remains a primary barrier to reliable deployment. To meet the demands of resource-constrained and high-stakes environments, practitioners must adopt multi-faceted evaluation frameworks that explicitly measure autonomy, resilience,

and memory governance. Bridging the gap between synthetic, controlled experiments and the noise of real-world financial workflows is essential for advancing the next generation of reliable, governable, and efficient memory-augmented agents.

References (APA 7th Edition)

Alqithami, S. (2025). Forgetful but faithful: A cognitive memory architecture and benchmark for privacy-aware generative agents. *arXiv.org, abs/2512.12856* null, . https://doi.org/10.48550/arxiv.2512.12856

Alshikh, W., Ali, M., Kennedy, B., & Mozolevskyi, D. (2025). Towards outcome-oriented, task-agnostic evaluation of AI agents. *arXiv.org, abs/2511.08242* null, . https://doi.org/10.48550/arxiv.2511.08242

Bandi, A., Kongari, B., Naguru, R., Pasnoor, S., & Vilipala, S. V. (2025). The rise of agentic AI: A review of definitions, frameworks, architectures, applications, evaluation metrics, and challenges. *Future Internet, 17* (9), 404-404. https://doi.org/10.3390/fi17090404

Bei, Y., Wei, T., Ning, X., Liu, Z., Lin, X., Zhu, Y., Hamann, H. F., He, J., & Tong, H. (2026). Mem-gallery: Benchmarking multimodal long-term conversational memory for mllm agents. https://doi.org/10.48550/arxiv.2601.03515

Bian, H., Yao, Z., Hu, S., Xu, Z., Zhang, S., Guo, Y., Yang, Z., Han, X., Wang, H., & Chen, R. (2026). Realmem: Benchmarking llms in real-world memory-driven interaction. https://doi.org/10.48550/arxiv.2601.06966

Bilal, M., & Nasrullah, N. (2025). Narrative memory based intelligent agents. *Deleted Journal, 5* (1), 104-112. https://doi.org/10.62019/abbdm.v5i1.298

Bousetouane, F. (2026). AI agents need memory control over more context. https://doi.org/10.48550/arxiv.2601.11653

Cherepanov, E., Kachaev, N., Zholus, A., Kovalev, A. K., & Panov, A. I. (2024). Unraveling the complexity of memory in rl agents: An approach for classification and evaluation. https://doi.org/10.48550/arxiv.2412.06531

Choppa, N. K. N. K. R. (2024). Contextual frameworks for agentic AI: Engineering adaptive memory and retrieval mechanisms. *Computer Fraud & Security*null, 395-406. https://doi.org/10.52710/cfs.747

Dann, C., Hofmann, K., & Nowozin, S. (2016). Memory lens: How much memory does an agent use?. *arXiv: Artificial Intelligence*null, .

Darshan, D., Varun, G., Rebecca, Q., & Peng, W. (2025). Memtrack: Evaluating long-term memory and state tracking in multi-platform dynamic agent environments. https://doi.org/10.48550/arxiv.2510.01353

Dian-xing, Z., wendong, L., Gang, L., Liu-chun, Y., & Sheng, L. (2025). Memory in large language models: Mechanisms, evaluation and evolution. https://doi.org/10.48550/arxiv.2509.18868

Ding, C., Si-min, N., Ke-hang, L., Peng, L., Xiang-ping, Z., Bo, T., Feiyu, X., & Zhiyu, L. (2025). Halumem: Evaluating hallucinations in memory systems of agents. https://doi.org/10.48550/arxiv.2511.03506

Dream — dynamic retention episodic architecture for memory. https://doi.org/10.5281/zenodo.17619916

Du, Y., Huang, W., Zheng, D., Wang, Z., Montella, S., Lapata, M., Wong, K., & Pan, J. Z. (2025). Rethinking memory in AI: Taxonomy, operations, topics, and future directions. *arXiv.org, abs/2505.00675* null, . https://doi.org/10.48550/arxiv.2505.00675

Hao-ran, S., & Shaoning, Z. (2025). Hierarchical memory for high-efficiency long-term reasoning in llm agents. https://doi.org/10.48550/arxiv.2507.22925

Haoran, X., Jiacong, H., Ke, Z., Lei, Y., Yuxin, T., Xinyuan, S., & Yiqun, D. (2025). Sedm: Scalable self-evolving distributed memory for agents. https://doi.org/10.48550/arxiv.2509.09498

Hu, S., Zhang, Z., Wei, Y., Han, X., Tang, Z., Wang, H., & Chen, R. (2026). Clonemem: Benchmarking long-term memory for AI clones. https://doi.org/10.48550/arxiv.2601.07023

Hu, Y., Wang, Y., & McAuley, J. (2025). Evaluating memory in llm agents via incremental multi-turn interactions. *arXiv.org, abs/2507.05257* null, . https://doi.org/10.48550/arxiv.2507.05257

Huang, R., Zhao, Z., Chen, S., Wu, X., & Zhao, J. L. (2025). Wealth-voyager: Navigating intelligentwealth management with a multi-agent framework. https://doi.org/10.1145/3766918.3766944

ibrahim, A. (2026). Graphmem: Self-evolving graph-based memory for production AI agents. https://doi.org/10.5281/zenodo.18131571

ibrahim, A. (2026). Graphmem: Self-evolving graph-based memory for production AI agents. https://doi.org/10.5281/zenodo.18131572

Lafuente-Mercado, R., Rus, D., & Rusch, T. K. (2025). Quantifying memory use in reinforcement learning with temporal range. *arXiv.org, abs/2512.06204* null, . https://doi.org/10.48550/arxiv.2512.06204

Li, X., ZHU, Z., Liu, S., Ma, Y., Zang, Y., Cao, Y., & Sun, A. (2026). Emembench: Interactive benchmarking of episodic memory for vlm agents. https://doi.org/10.48550/arxiv.2601.16690

Li, Y. E., Liu, J., Wang, Y., Wu, Y., & Xu, M. (2026). Bmam: Brain-inspired multi-agent memory framework. https://doi.org/10.48550/arxiv.2601.20465

Liang, S., Cao, P., Zhao, J., Teng, W., Zhao, J., & Liu, K. (2026). Learning how to remember: A meta-cognitive management method for structured and transferable agent memory. https://doi.org/10.48550/arxiv.2601.07470

Logan, J. (2026). Continuum memory architectures for long-horizon llm agents. https://doi.org/10.48550/arxiv.2601.09913

Memoriesdb: A temporal-semantic-relational database for long-term agent memory. https://doi.org/10.5281/zenodo.17469799

Memory management and contextual consistency for long-running low-code agents. https://doi.org/10.48550/arxiv.2509.25250

Mohammadi, M., Li, Y., Lo, J., & Yip, W. (2025). Evaluation and benchmarking of llm agents: A survey. https://doi.org/10.1145/3711896.3736570

Myakala, P. K. (2025). Beyond accuracy: A multi-faceted evaluation framework for real-world AI agents. https://doi.org/10.5281/zenodo.14880715

Rasmussen, P., Paliychuk, P., Beauvais, T., Ryan, J., & Chalef, D. (2025). Zep: A temporal knowledge graph architecture for agent memory. https://doi.org/10.48550/arxiv.2501.13956

Shen, Y., Li, K., Zhou, W., & Hu, S. (2026). Mem2actbench: A benchmark for evaluating long-term memory utilization in task-oriented autonomous agents. https://doi.org/10.48550/arxiv.2601.19935

Shen, Y., Pei, D., Guo, Y., Guo, Y., Zhang, Z., Jia, Q., Zhou, J., & Zhai, G. (2026). Evolmem: A cognitive-driven benchmark for multi-session dialogue memory. https://doi.org/10.48550/arxiv.2601.03543

Shian, J., Ziyang, H., Xin-bo, W., Haofei, Z., & Mingli, S. (2025). Pisa: A pragmatic psych-inspired unified memory system for enhanced AI agency. https://doi.org/10.48550/arxiv.2510.15966

Sinha, U. (2025). Autonomously transacting agents: A new paradigm for AI in finance. *Journal of computer science and technology studies*null, . https://doi.org/10.32996/jcsts.2025.7.12.58

Souza, D., & Machado, P. (2026). Toward architecture-aware evaluation metrics for llm agents. https://doi.org/10.48550/arxiv.2601.19583

Sun, R., Bai, Z., Zhang, W., Zhang, Y., Zhao, L., Sun, S., & Qiu, Z. (2025). Finresearchbench: A logic tree based agent-as-a-judge evaluation framework for financial research agents. https://doi.org/10.48550/arxiv.2507.16248

Wang, P., Tian, M., Li, J., Liang, Y., Wang, Y., Chen, Q., Wang, T., Lu, Z., Ma, J., Jiang, Y., & Zhou, W. (2025). O-mem: Omni memory system for personalized, long horizon, self-evolving agents. *arXiv.org, abs/2511.13593* null, . https://doi.org/10.48550/arxiv.2511.13593

Wei, T., Sachdeva, N., Coleman, B., He, Z., Bei, Y., Ning, X., Ai, M., Li, Y., He, J., Chi, E., Wang, C., Chen, S., Pereira, F., Kang, W., & Cheng, D. Z. (2025). Evo-memory: Benchmarking llm agent test-time learning with self-evolving memory. *arXiv.org, abs/2511.20857* null, . https://doi.org/10.48550/arxiv.2511.20857

Xiong, Z., Lin, Y., Xie, W., He, P., Tang, J., Lakkaraju, H., & Xiang, Z. (2025). How memory management impacts llm agents: An empirical study of experience-following behavior. *arXiv.org, abs/2505.16067* null, . https://doi.org/10.48550/arxiv.2505.16067

Xu, B. (2026). AI agent systems: Architectures, applications, and evaluation. https://doi.org/10.48550/arxiv.2601.01743

Yan, B. Y., Li, C., Qian, H., Lu, S., & Liu, Z. (2025). General agentic memory via deep research. *arXiv.org, abs/2511.18423* null, . https://doi.org/10.48550/arxiv.2511.18423

Yan, S., Yang, X., Huang, Z., Nie, E., Ding, Z., Li, Z., Ma, X., Schutze, H., Tresp, V., & Ma, Y. (2025). Memory-r1: Enhancing large language model agents to manage and utilize memories via reinforcement learning. *arXiv.org, abs/2508.19828* null, . https://doi.org/10.48550/arxiv.2508.19828

Yang, X., Li, L., Zhou, H., Zhu, T., Qu, X., Fan, Y., Wei, Q., Ye, R., Kang, L., Qin, Y., Kou, Z., Liu, D., Li, Q., Ding, N., Chen, S., & Shao, J. (2026). Toward efficient agents: Memory, tool learning, and planning. https://doi.org/10.48550/arxiv.2601.14192

Yu, W., Ryuichi, T., Zhiqi, L., Yuzhen, M., Yuanzhe, H., Julian, M., & Xiao-jian, W. (2025). Mem-α: Learning memory construction via reinforcement learning. https://doi.org/10.48550/arxiv.2509.25911

Yu, Y., Yao, L., Xie, Y., Tan, Q., Feng, J., Li, Y., & Wu, L. (2026). Agentic memory: Learning unified long-term and short-term memory management for large language model agents. https://doi.org/10.48550/arxiv.2601.01885

Zhang, Z., Bo, X., Ma, C., Li, R., Chen, X., Dai, Q., Zhu, J., Dong, Z., & Wen, J. (2024). A survey on the memory mechanism of large language model based agents. *arXiv.org, abs/2404.13501* null, . https://doi.org/10.48550/arxiv.2404.13501

Zhang, Z., Dai, Q., Li, R., Bo, X., Chen, X., & Dong, Z. (2025). Learn to memorize: Optimizing llm-based agents with adaptive memory framework. *arXiv.org, abs/2508.16629* null, . https://doi.org/10.48550/arxiv.2508.16629

Zhang, Z., Zhang, Y., Tan, H., Li, R., & Chen, X. (2025). Explicit v.s. Implicit memory: Exploring multi-hop complex reasoning over personalized information. *arXiv.org, abs/2508.13250* null, . https://doi.org/10.48550/arxiv.2508.13250